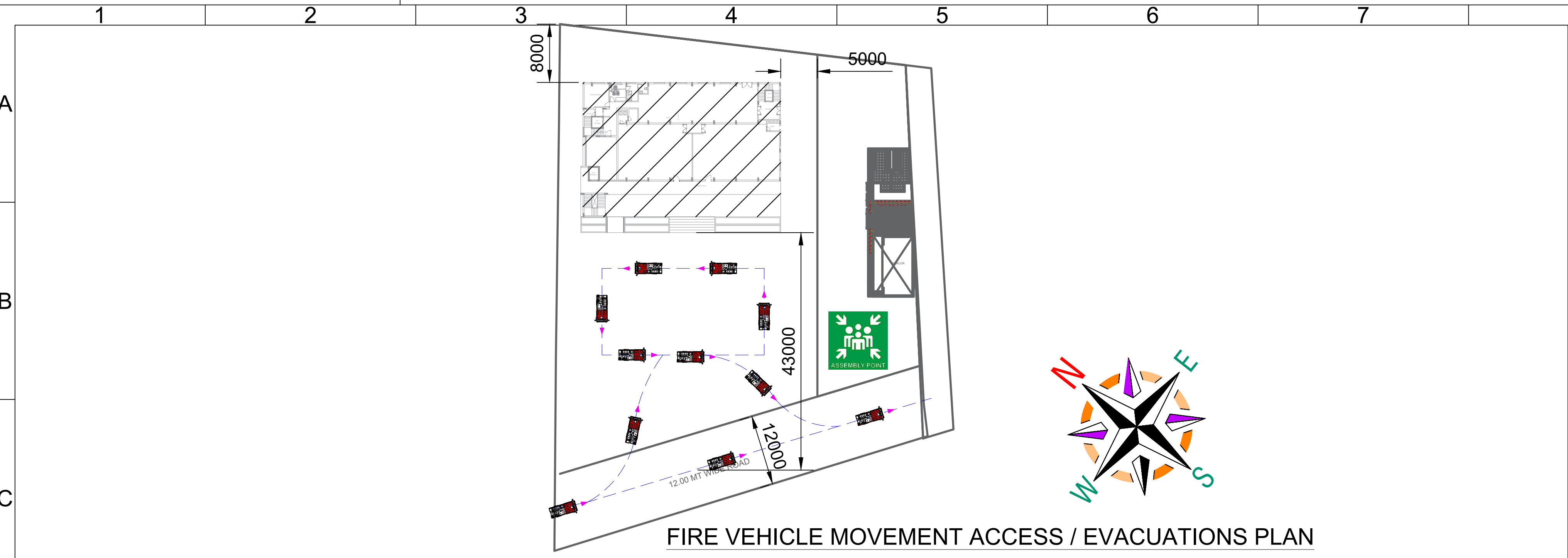




1. HYDRANT SYSTEM:

1_ON/OFF switches located near the hose reel hose or hydrant outlet, at each floor the main FirePump at the underground water tank with a capacity to discharge 900 liters per minute at 3.5 bar. pressure as measured at the terrace level should be installed.

2_The Riser for the buildings exceeding 18 meters and above 18 meters height should not be of less than 100 mm. internal diameter. The riser should be connected to the bottom of the terrace tank with a stop valve and a NRV to act as a Down – comer.

3_One riser is required for every 1000sq. meters floor area and if the building is divided into, two or more parts then each part should have a separate riser with all the fittings at floor level. Each floor should have one hydrant outlet with a coupling for attaching a 63 mm.dia hose & 25 mm.bore Hose–reel hose with 8 mm. SS. Shut–off nozzle at each floor landing. The length of the hose reel hose should be enough to reach the farthest corner of the floor. Hose–box with 15 meters long 63mm. dia. hose and 12.5 mm. bore nozzle at alternate floors. The Hose–reel hose should be coupled to the Riser.

Fire–service inlet should be installed at a point near the entry to the premises where a Fire service vehicle can approach easily. A permanent hydrant point comprising of 63 mm.dia size 2nos of hydrant valves should be installed at the terrace level.

Overhead tank Refilling bypass connection should be done at the terrace level.

4_The Overhead tank shall be of a capacity as per NBC 2016 guide line. The Underground tank capacity as per NBC 2016 guide line.

5_The distance between two hydrant is 45meters in yard.

2. FIRE LIFT :

The Fire–Lift and all the lifts should have a provision to ground automatically in case of electricity failure. Each building should have at least one lift as a Fire–Lift and if the building is divided into two or more parts then each should have a Fire–Lift. Lift–well should have blowers to pressurize the lift–well so connected that it will automatically operate when alarm call point is operate, so that it prevents the lift well getting smoke logged.

3. FIRE ALARM:

Fire alarm call point to be installed at each floor with sounders capable of being heard all throughout the building.

4. FIRE EXTINGUISHERS

One Carbon Dioxide (CO2) type extinguisher of 4.5 kg. with ISI mark. and one extinguisher of 5kg/6KG. Dry Chemical Powder (DCP) type extinguisher with ISI mark to be installed on each floor in case of commercial building.

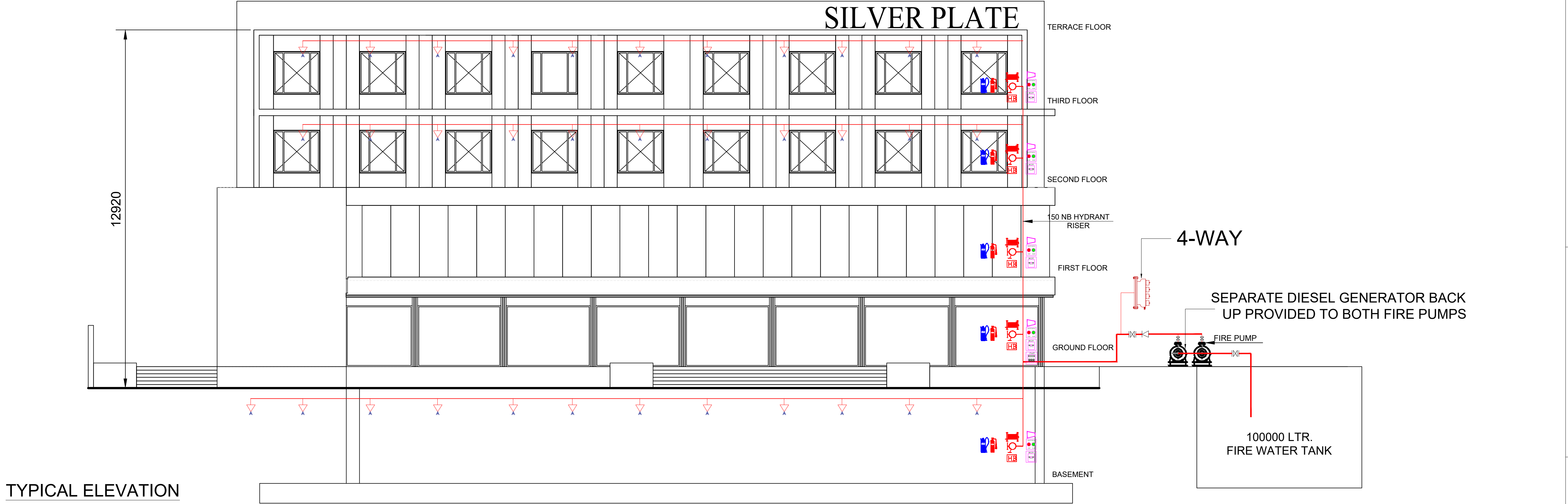
One Carbon Dioxide (CO2) type extinguisher of 4.5 kg. with ISI mark. OR Two Carbon Dioxide (Co2) type extinguisher of 2 kg. with ISI mark capacity on alternate floors in case of residential building.

If the building is divided into two or more parts then each part should have these extinguishers installed.



KEY PLAN

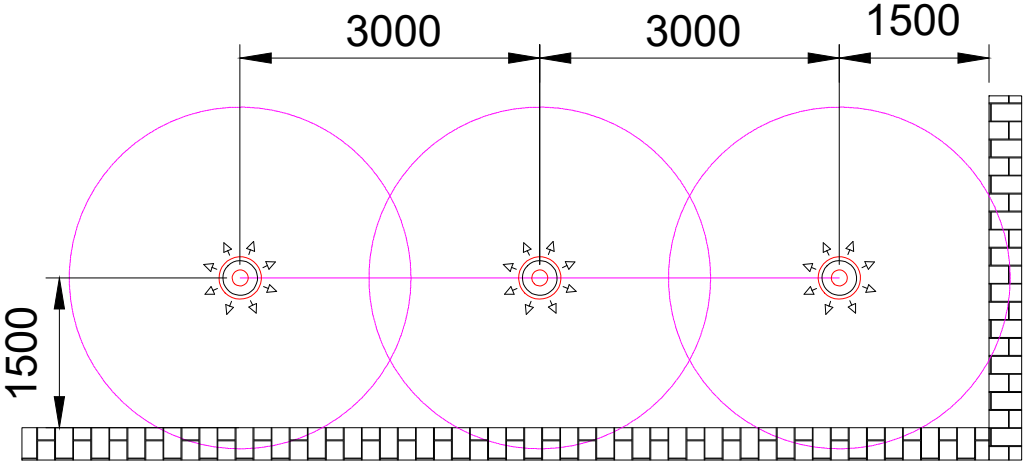
LEGEND:-		
SR.NO	SYMBOL	DESCRIPTION
1		100NB PIPE
2		80NB PIPE
3		25NB PIPE
4		32NB PIPE
5		40NB PIPE
6		50NB PIPE
7		FIRE HYDRANT VALVE
8		HOSE REEL
9		HOSE BOX
10		FAP – FIRE ALARM PANEL
11		FIRE MANUAL CALL POINT
12		FIRE PUMP ON-OFF SWITCH
13		FIRE HOOTER
14		FIRE PUMP
15		BPV – ISOLATION VALVE
16		NRV – NON RETURN VALVE
17		6KG ABC (DCP) TYPE FIRE EXTINGUISHER
18		4.5KG CO2 TYPE FIRE EXTINGUISHER
19		4WAY FIRE BRIGADE INLET
20		FIRE WATER SPRINKLER
		SMOKE DETECTOR
21		SMOKE DETECTOR



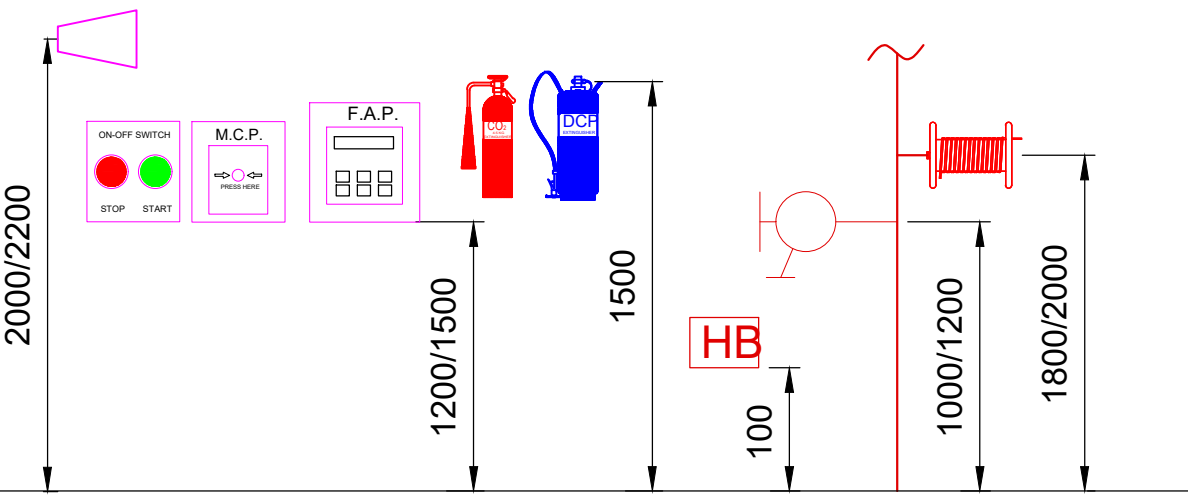
DETAILS OF FIRE FIGHTING PUMPS			
SR. NO	DESCRIPTION	CAPACITY	HEAD (MWC)
1	END SUCTION MONO BLOCK PUMP	1620 LPM	65 M
2	END SUCTION MONO BLOCK PUMP	180 LPM	65 M

FIRE WATER STORAGE TANK

UNDERGROUND TANK	100000 LTRS
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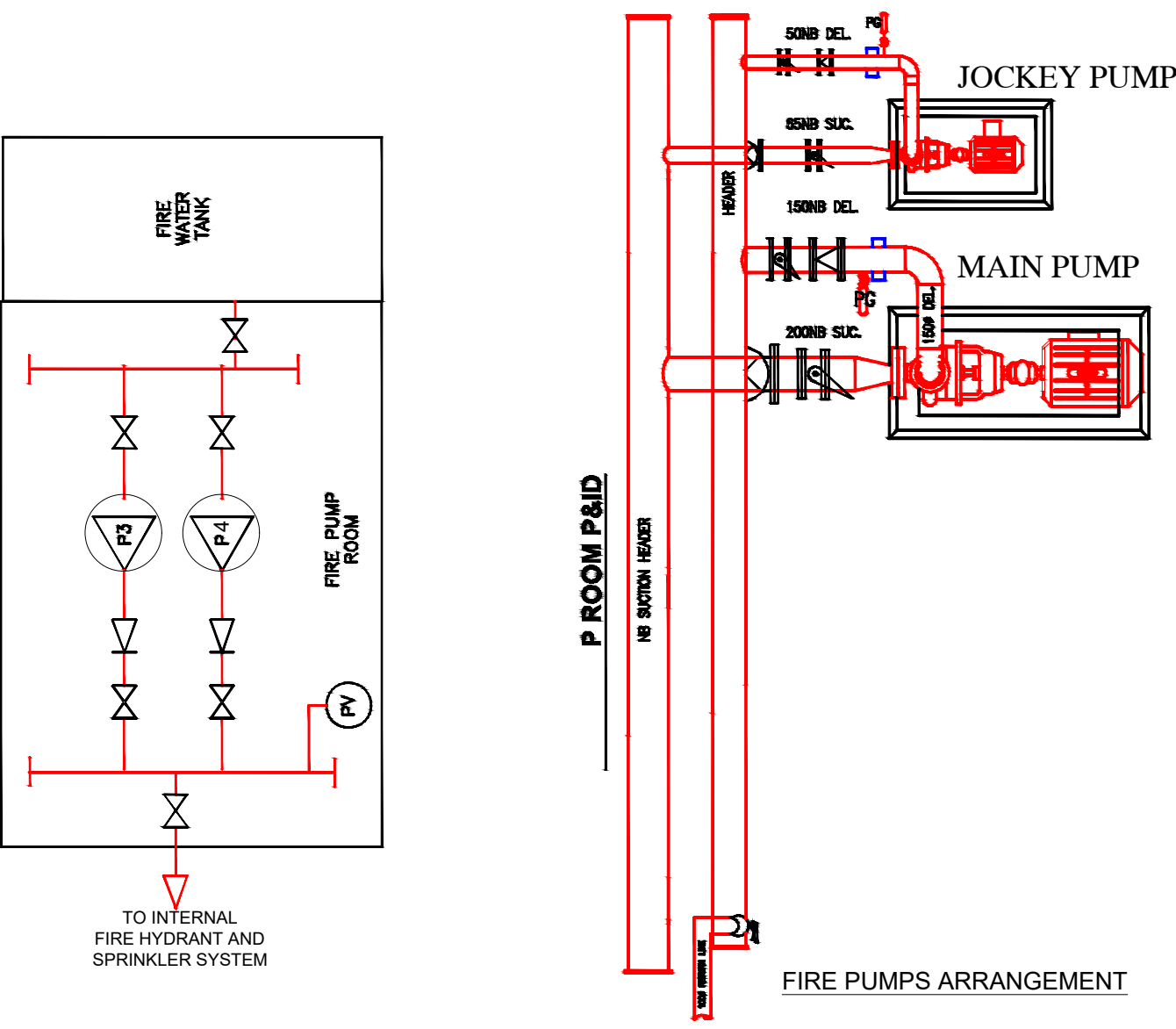
DISTANCE BETWEEN TWO SPRINKLER AND DISTANCE FROM WALL



TYPICAL DETAIL OF FIRE EQUIPMENTS MOUNTING HEIGHTS.

SR. NO.	AREA TABLE	IN SQ.MTS.
1	BASEMENT FLOOR	174.45 SQ.MTS.
2	GROUND FLOOR	817.89 SQ.MTS.
3	FIRST FLOOR	695.79 SQ.MTS.
4	SECOND FLOOR	642.88 SQ.MTS.
5	THIRD FLOOR	532.82 SQ.MTS.
	TOTAL	2863.83 SQ.MTS.

ITEMS QUANTITY (MINIMUM REQUIRED QUANTITY)			
SR.NO	DESCRIPTION	QTY	UNIT
1	FIRE HYDRANT VALVE, IS:5290	05	NOS.
2	HOSE BOX, MS sheet.	05	NOS.
3	HOSE PIPE, (15Mtrs.) IS: 636	05	NOS.
4	BRANCH PIPE, IS 903,	05	NOS.
5	HOSE REEL, 30mtr (For Internal Staircase)	05	NOS.
6	4 WAY FIRE BRIGADE INLET (FBI), IS: 904.	01	NO.
7	4.5KG CO2 TYPE FIRE EXTINGUISHER	05	NOS.
8	6KG ABC (DCP) TYPE FIRE EXTINGUISHER	05	NOS.
9	FIRE ALARM PANEL	01	NOS.
10	MANUAL CALL POINT	05	NOS.
11	FIRE HOOTER	05	NOS.
12	SPRINKLER	67	NOS.
13	SMOKE DETECTOR	76	NOS.



OCCUPANT LOAD

SR.NO	DESCRIPTION		
a	FLOOR AREA = LENGTH x WIDTH		441.32 M ²
b	OCCUPANT LOAD PER FLOOR = AREA/LOAD FACTOR	441.32 / 1.40	315.22 M ² /PERSON
c	TOTAL NOS. OF FLOOR PER BLOCK.(B+G+3)	4 x 315	1260 PERSON
d	TOTAL NOS. OF BLOCKS ARE 1 NO.	1260 X 1	1260 PERSON

NUMBER OF EXIT

a	EXIT CAPACITY (OCCUPANT LOAD PER FLOOR X CAP.FACTOR)	315 x 10	3150MM (3.15M)
b	MINIMUM REQUIRED OF EXIT. (EXIT CAPACITY / STAIR CASE)	3.15 / 1.5	2 NOS.
c	MINIMUM HEIGHT AND WIDTH	H–2000MM	W–1500M

TITLE : FIRE PROTECTION SYSTEM FOR MIXED OCCUPANCY
HOTEL SILVER PLATE,
AT-VILLAGE DASHRATH,
VADODARA

DWG. NO: SSF-26-27-13		AP.N: XXX	
Date 20/05/26	Draw	Size A1	Scale 240
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